

Muhammad Talha

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Summary

Computer Science graduate with practical experience in machine learning, backend development, and AI applications. Skilled in building end-to-end ML pipelines, deploying models with FastAPI and Docker. Interested in an entry-level Machine Learning, Data Science, or Software Engineering role.

Technical Skills

- **Programming:** Python, SQL, C/C++, HTML, CSS
- **Machine Learning :** Scikit-learn, PyTorch, Transformers, NLP
- **Backend:** FastAPI, REST APIs, Pydantic
- **Data:** Pandas, NumPy, Feature Engineering, EDA
- **Tools:** Docker, Git, Linux, VS Code, Jupyter Notebook, Google Colab

Experience

- **Data Science Trainee** June 2026 – July 2026
Machine Learning 1 Onsite
 - Worked on real-world machine learning projects using Python and Scikit-learn.
 - Performed data preprocessing, exploratory data analysis (EDA), and feature engineering.
 - Built, trained, and evaluated classification models using cross-validation and hyperparameter tuning.
 - Collaborated using Git while following machine learning development workflows.
 - Gained hands-on experience in developing machine learning pipelines.

Projects

- **Next-Gen AI Desktop Assistant (Final Year Project)** Gen AI
 - Developed an offline AI desktop assistant supporting voice and text interactions using local LLMs.
 - Implemented modular task execution, file management, application and system control, and conversational workflows.
 - Integrated long-term memory with ChromaDB to enable context-aware conversations while preserving user privacy.**Tools:** Python, Ollama, Gemma 3, ChromaDB, PySide6, WhisperFlow, NLTK
- **Insurance Premium Category Predictor** ML Engineering / Backend
 - Built an end-to-end machine learning pipeline for insurance premium category prediction.
 - Developed and deployed a REST API using FastAPI and Docker for scalable inference.
 - Implemented structured logging, preprocessing pipelines, and error handling for production readiness.**Tools:** FastAPI, Docker, Scikit-learn, Python, Linux
- **Multilingual Sentiment Analysis (Urdu & English)** NLP / Transformers
 - Fine-tuned a multilingual BERT (mBERT) model for Urdu–English sentiment classification.
 - Built a custom preprocessing and tokenization pipeline to handle code-switched text.
 - Trained and evaluated transformer models using the Hugging Face ecosystem and PyTorch.**Tools:** Hugging Face Transformers, PyTorch, mBERT

Research Experience

- **Beyond the Cloud: Offline AI Desktop Assistant** 2026
Undergraduate Research (Final Year Project)
 - Co-authored a research paper on designing an offline-first AI desktop assistant using local large language models.
 - Proposed a privacy-preserving architecture integrating local LLM inference, speech recognition, and modular command execution.
 - Contributed to literature review, system design, implementation, experimental evaluation, and technical documentation.**Keywords:** Local LLMs, Edge AI, NLP, Speech Recognition, Privacy-Preserving AI

Education

- **University of Engineering and Technology, Lahore** 2022 – 2026
Bachelor of Science in Computer Science

Certifications

- **Machine Learning Specialization** – Stanford / DeepLearning.AI (Coursera) 2025
- **Intermediate SQL** – DataCamp 2025